



Probability · Worksheet 1

Statistics Chapter 5: Probability. Answer all questions. Show your working clearly. Total: 82 marks. Time: 100 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are four sections, one for each topic in Chapter 5. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- Give probabilities as exact fractions or decimals (3 significant figures unless exact).
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Venn diagrams and set notation	19	
B Mutually exclusive and independent events	19	
C Tree diagrams	20	
D Conditional probability	19	
Roar-level challenge	5	
Total	82	

Section A Venn diagrams and set notation

19 marks

1. $P(A) = 0.5$, $P(B) = 0.4$ and $P(A \cap B) = 0.2$.
Find (a) $P(A \cup B)$, (b) $P((A \cup B)')$.

[2]

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2. In a class of 30 students, 18 play football and 22 play tennis. Every student plays at least one of these sports. [3]

- (a) Find the number of students who play both sports.
(b) A student is chosen at random. Find the probability that the student plays football but not tennis.

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3. $P(A) = 0.45$, $P(B) = 0.5$ and $P(A \cup B) = 0.8$. [3]
Find (a) $P(A \cap B)$, (b) $P(A' \cap B)$.

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4. 50 people were asked which of coffee (C), tea (T) and cocoa (K) they drink. [4]
20 drink coffee, 15 drink tea and 12 drink cocoa. 6 drink coffee and tea, 4 drink coffee and cocoa, 3 drink tea and cocoa, and 2 drink all three.

- (a) Draw a Venn diagram to represent this information.
(b) Find the number of people who drink none of the three.
(c) A person is chosen at random. Find the probability that they drink exactly one of the three.

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5. The events A , B and C are such that $P(A) = 0.4$, $P(B) = 0.35$, $P(C) = 0.3$.
 A and C are mutually exclusive, $P(A \cap B) = 0.15$ and $P(B \cap C) = 0.1$.

(a) Draw a Venn diagram showing the probabilities for all regions.

[4]

(b) Find $P(A \cup C)$.

[1]

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(c) Find $P(B' \cap C')$.

[2]

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(Total for Question 5 is 7 marks)



Section B Mutually exclusive and independent events

19 marks

6. A and B are mutually exclusive with $P(A) = 0.35$ and $P(B) = 0.4$. [2]

Write down (a) $P(A \cap B)$, (b) $P(A \cup B)$.

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7. A and B are independent with $P(A) = 0.25$ and $P(B) = 0.6$. [3]

Find (a) $P(A \cap B)$, (b) $P(A \cup B)$, (c) $P(A' \cap B')$.

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8. $P(A) = 0.4$, $P(B) = 0.25$ and $P(A \cap B) = 0.1$. [3]

(a) Show that A and B are independent.

(b) State, with a reason, whether A and B are mutually exclusive.

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9. 100 adults were asked whether they smoke and whether they exercise regularly.

[4]

	Exercise regularly	Do not exercise regularly
Smoker	12	18
Non-smoker	48	22

One adult is chosen at random. S is the event “the adult smokes” and E is the event “the adult exercises regularly”.

(a) Find $P(S)$ and $P(S \cap E)$.

(b) Determine whether S and E are independent. Show your working.

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10. $P(A) = 0.4$, $P(B) = x$ and $P(A \cup B) = 0.7$.

(a) Find x if A and B are mutually exclusive.

[2]

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(b) Find x if A and B are independent.

[3]

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(c) In the case where A and B are independent, find $P(A' \cap B)$.

[2]

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(Total for Question 10 is 7 marks)



Section C Tree diagrams

20 marks

11. An archer hits the target with probability 0.9 on each shot, independently. She takes two shots. [3]

- (a) Draw a tree diagram.
- (b) Find the probability she hits with both shots.
- (c) Find the probability she misses at least once.

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12. The probability that a bus is late on any day is 0.4, independently of other days. [3]
Find the probability that the bus is late on exactly one of two days.

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13. A bag contains 4 red and 6 blue counters. Two counters are taken at random without replacement. [3]
Find the probability that (a) both are red, (b) at least one is blue.

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14. Three independent machines each work on a given day with probability 0.7. [4]
Find the probability that on a given day (a) exactly two machines work, (b) at least one machine works.

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15. A bag contains 5 red, 4 blue and 3 green counters. Two counters are taken at random, one after the other, without replacement.

(a) Draw a tree diagram to show the possible outcomes, with probabilities.

[2]

(b) Find the probability that both counters are the same colour.

[2]

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(c) Find the probability that at least one counter is red.

[2]

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(d) Explain why the number of red counters taken cannot be modelled by $B(2, \frac{5}{12})$.

[1]

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(Total for Question 15 is 7 marks)



Section D Conditional probability

19 marks

16. $P(A \cap B) = 0.12$ and $P(B) = 0.4$. Find $P(A | B)$. [2]

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17. $P(A) = 0.5$, $P(B) = 0.5$ and $P(B | A) = 0.4$. [3]
Find (a) $P(A \cap B)$, (b) $P(A | B)$.

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18. In a year group of 60 students, 38 study French, 30 study Spanish and 16 study neither. [3]
A student who studies French is chosen at random. Find the probability that this student also studies Spanish.

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19. 10% of the components made by a factory are faulty. A test correctly identifies a faulty component with probability 0.9, but wrongly identifies a good component as faulty with probability 0.03. [4]

- (a) Find the probability that a randomly chosen component is identified as faulty.
(b) Given that a component is identified as faulty, find the probability that it is actually faulty.

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20. The events A and B are such that $P(A) = 0.4$, $P(B) = 0.35$ and $P(A \cup B) = 0.6$. [1]
- (a) Find $P(A \cap B)$.

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- (b) Find $P(B | A)$. [2]

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- (c) Find $P(A | B')$. [2]

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- (d) Determine, with a reason, whether A and B are independent. [2]

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(Total for Question 20 is 7 marks)



ROAR-LEVEL CHALLENGE

5 marks

21. Rex walks to school with probability 0.5; otherwise he takes the bus. If he walks, the probability he is late is 0.4. If he takes the bus, the probability he is late is 0.2. [5]
- (a) Find the probability that Rex is late. (b) Given that Rex is late, find the probability that he walked.
- (c) Determine whether the events “walks” and “is late” are independent.

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END OF WORKSHEET

Total: 82 marks